
VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION OF VISION–LANGUAGE MODELS

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ABSTRACT

Vision–Language Models (VLMs) constitute a promising class of zero-shot goal-conditioned value functions, yet their frozen pre-trained representations impose two structural constraints: limited generalization under distribution shift and weak temporal reasoning over trajectories. We introduce VITA, a zero-shot value function learning method that addresses both constraints through test-time adaptation. At inference time, a lightweight adaptation module is updated by a single gradient step on a meta-learned self-supervised loss, with the meta-objective enforcing that each such update strictly improves value estimation. Because the module is updated sequentially along a trajectory, VITA accumulates observation history directly in its parameters, providing an implicit memory that supplies the temporal context absent from frozen representations. To suppress shortcut learning, we introduce a dissimilarity-based sampling strategy that, at training time, selects the most semantically diverse sub-trajectory segments for adaptation. On real-world robotic manipulation tasks, VITA trained in a single environment generalizes zero-shot to out-of-distribution tasks, environments, and embodiments, exceeding the state-of-the-art zero-shot method built on autoregressive VLMs. We further show that the zero-shot value estimates produced by VITA serve as a dense reward-shaping signal for offline reinforcement learning, yielding multi-task policies on the Meta-World benchmark that surpass those trained with the simulator’s fuzzy-logic dense rewards.

1 INTRODUCTION

Vision–Language Models (VLMs) attain strong generalization across heterogeneous tasks and domains by optimizing over large-scale, unstructured web corpora without human supervision (Radford et al., 2021; Alayrac et al., 2022). Generalist control policies, by contrast, have not matched this level of transfer: despite substantial progress in robotics (Brohan et al., 2023; Kim et al., 2024) and 3D virtual environments (Reed et al., 2022; SIMA Team et al., 2024), state-of-the-art methods remain dependent on expert demonstrations for supervision (Ho & Ermon, 2016). World models (Ha & Schmidhuber, 2018; Hafner et al., 2021; Matsuo et al., 2022) offer one route around this dependency, allowing agents to be trained within learned simulators, with instantiations spanning robotics (Yang et al., 2023; Agarwal et al., 2025), autonomous driving (Hu et al., 2024), and video games (Bruce et al., 2024). Even granting that such models can synthesize realistic visual trajectories in diverse, open-ended environments (Hughes et al., 2024; Silver & Sutton, 2025), a central problem is left unresolved: *how can agents learn from video at scale without recourse to human supervision?*

One line of research learns policies directly from expert visual trajectories by inferring the latent action underlying each visual transition (Edwards et al., 2019; Baker et al., 2022; Schmidt & Jiang, 2024; Zhang et al., 2025a). These inferred latent actions can, in principle, be decoded into executable controls at deployment, though reliably learning this mapping remains difficult in practice. A parallel line instead learns a universal *goal-conditioned value function* and uses it as a zero-shot signal—for reward shaping in reinforcement learning (RL) and for supervision in imitation learning, respectively (Chen et al., 2021; Ma et al., 2022). Under this formulation, goal-conditioned value estimation reduces to a regression problem: predict the fraction of a task an agent has completed, conditioned on a visual observation and a natural-language task description (Lee et al., 2021a; Ma et al., 2023b; 2024; Dashora et al., 2025).

Prior work applies pre-trained contrastive VLMs (Radford et al., 2021) to zero-shot reward shaping, scoring progress by the similarity between the task description and each visual observation within a shared multimodal embedding space (Ma et al., 2023b; Baumli et al., 2023; Rocamonde et al., 2024). Because these scores are computed frame-by-frame, they omit the temporal context required to disambiguate visually near-identical states that occur at distinct task stages (e.g., folding versus unfolding a shirt). Autoregressive VLMs (Alayrac et al., 2022) instead inject temporal context by conditioning on the full visual trajectory in the prompt, but the chronologically ordered pre-training corpora induce a systematic bias toward monotonically increasing progress predictions (Ma et al., 2024). Crucially, both architectures perform zero-shot prediction on top of frozen pre-trained representations (Ma et al., 2023b; 2024; Baumli et al., 2023; Rocamonde et al., 2024; Du et al., 2023), and this reliance jointly bounds their generalization and their temporal reasoning (Pătrăucean et al., 2023). To recover generalizable value estimation, recent work resorts to large-scale pre-training (Alakuijala et al., 2024; Ma et al., 2023b), domain-specific fine-tuning (Zhang et al., 2025b), or demonstration-based reward specification (Sontakke et al., 2023).

This work introduces VITA, a zero-shot value function learning method that strengthens both generalization and temporal reasoning by means of test-time adaptation, and that surpasses the state-of-the-art zero-shot approach (Ma et al., 2024) on real-world robotic manipulation tasks. In contrast to methods that acquire generalization through large-scale pre-training or expert demonstrations, VITA adapts online to the semantic and temporal context of each trajectory. At inference, a lightweight adaptation module is updated—at negligible computational cost—by one gradient step on a meta-learned self-supervised loss (Sun et al., 2024), where the meta-objective is defined so that each such update improves value estimation (Finn et al., 2017). Applying these updates sequentially along a test trajectory causes the estimator to encode trajectory history in its parameters (Sun et al., 2024), which directly targets the temporal-reasoning limitation identified above. To suppress shortcut learning (Geirhos et al., 2020), we further introduce a dissimilarity-based sampling strategy that biases adaptation toward semantic cues, an effect we validate empirically. We evaluate VITA along three core axes of value function estimation: generalization under distribution shift, discrimination between expert and non-expert trajectories, and dense reward shaping for offline RL (Levine et al., 2020).

Our contributions are as follows:

- We propose VITA, a test-time adaptation mechanism that augments the generalization and temporal reasoning of contrastive VLMs for zero-shot value function estimation, while requiring neither task-specific demonstrations nor large-scale pre-training.
- We show that VITA, trained in a single environment, generalizes zero-shot to out-of-distribution tasks, environments, and embodiments in robotic manipulation, exceeding the state-of-the-art zero-shot method (Ma et al., 2024).
- We show that the zero-shot value estimates produced by VITA, used as a dense reward-shaping signal, yield offline RL policies on the Meta-World multi-task benchmark (MT10) that outperform those trained with the simulator’s fuzzy-logic dense rewards (Yu et al., 2020).

2 PRELIMINARIES

2.1 GOAL-CONDITIONED VALUE FUNCTIONS

For video trajectories, task progress estimation is equivalent to goal-conditioned value function estimation under a reward that encodes task completion. We therefore cast the problem as regressing the degree of task completion from vision–language representations. Formally, the value function $V : \mathcal{O} \times \mathcal{G} \rightarrow [0, 1]$ maps an observation $o_t \in \mathcal{O}$ and a goal specification $g \in \mathcal{G}$ to a scalar estimate of progress toward goal completion, with the endpoints fixed by convention: $V(o_t; g) = 0$ at the task’s start and $V(o_t; g) = 1$ at its completion. Following standard practice, we identify progress with temporal position in expert demonstrations, under the assumption that such trajectories are monotonically increasing in progress toward the goal (Lee et al., 2021a; Ma et al., 2024; Dashora et al., 2025). Concretely, for an expert trajectory $\tau = (o_1, \dots, o_T) \sim \pi_E$ of length T , the progress target is the normalized timestep index $V^{\pi_E}(o_t; g) = \frac{t}{T}$, which supplies the supervision signal used to fit the goal-conditioned value function V .

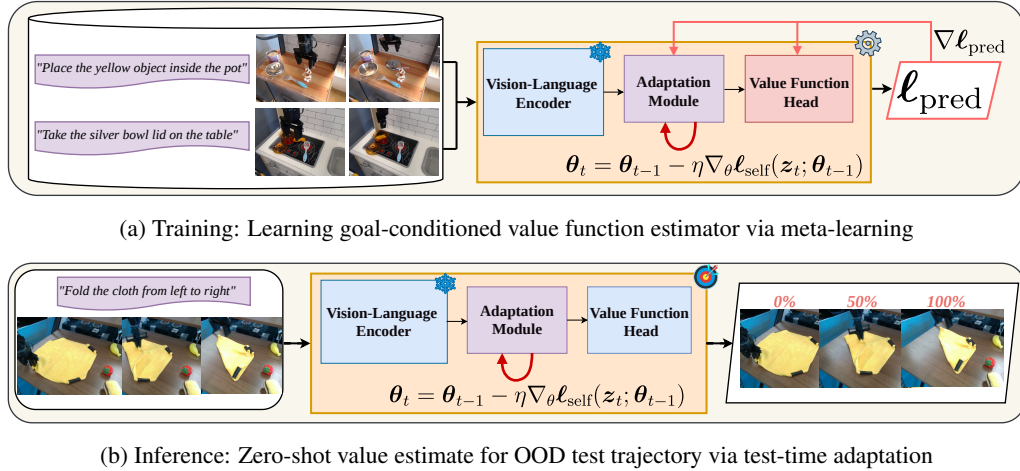


Figure 1: **Overview of VITA.** At training time, VITA fits a goal-conditioned value function by gradient-based meta-learning; at inference time, it attains zero-shot generalization to out-of-distribution trajectories through sequential test-time adaptation of a lightweight module.

2.2 TEST-TIME TRAINING FOR SEQUENCE MODELING

Test-time training (TTT) (Sun et al., 2020) is a test-time adaptation procedure that reformulates inference as a self-supervised optimization (Goodfellow et al., 2016), updating model parameters on each test instance without access to labels. Although originally formulated for static image classification (Sun et al., 2020), subsequent work extends TTT to sequence modeling (Sun et al., 2024; Wang et al., 2023), in which a parametric adaptation module f_{adapt} is updated at every timestep by a gradient step on a self-supervised loss. Whereas conventional sequence architectures store temporal history in activation state—recurrent hidden states (Hochreiter & Schmidhuber, 1997; Cho et al., 2014) or cached key–value pairs (Vaswani et al., 2017)—TTT instead writes this history into the *parameters* of f_{adapt} through the sequence of inference-time updates. The parameters of f_{adapt} thus act as an implicit memory that encodes temporal history while preserving temporal order. Sun et al. (2024) additionally propose to meta-learn the self-supervised task under the gradient-based meta-learning paradigm (Finn et al., 2017): rather than minimizing the self-supervised loss directly, in this setting (Sun et al., 2024) the loss is parameterized by learnable linear projections that are optimized so that a test-time update reduces the downstream prediction loss. We adapt this construction to goal-conditioned value function learning, capturing temporal context via sequential updates and meta-learning the self-supervised task to improve value estimation instead of fixing it a priori.

3 VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION

3.1 MODEL ARCHITECTURE

Our goal-conditioned value function estimator is composed of three modules: (1) a frozen multimodal encoder that maps a visual trajectory and its task description to a sequence of joint vision–language representations; (2) an adaptation module updated at test time by a self-supervised loss whose parameterization is meta-learned to improve value estimation; and (3) a regression head that maps the adapted representation to a scalar value estimate.

3.1.1 MULTIMODAL INPUT REPRESENTATION

We employ a frozen contrastive vision–language encoder, CLIP (Radford et al., 2021), to embed visual observations and goal descriptions. Given a visual trajectory $\tau = (o_1, \dots, o_T)$ and a language task description g , we form, at each timestep t , the joint multimodal representation by concatenating the two modality embeddings, yielding the sequence $(z_1, z_2, \dots, z_T)$ with $z_t = [\phi_v(o_t); \phi_g(g)] \in \mathbb{R}^{2d}$, where ϕ_v and ϕ_g denote the visual and language encoders, respectively. Because CLIP is

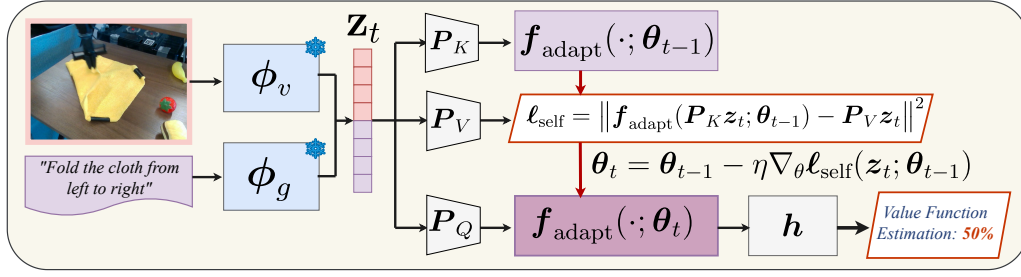


Figure 2: **Test-time Adaptation.** At inference, for each timestep t the adaptation module f_{adapt} undergoes one gradient step on the meta-learned self-supervised loss ℓ_{self} ; iterating this update along the trajectory accumulates temporal history in the module’s parameters.

pre-trained with a contrastive objective that aligns paired image–text inputs in a shared embedding space (Radford et al., 2021), observations that are semantically nearer to goal completion tend to lie closer to the goal-description embedding, which grounds the value estimate in semantic similarity.

3.1.2 TEST-TIME ADAPTATION

To align the multimodal representations with both semantic and temporal context, we introduce an adaptation module f_{adapt} instantiating the test-time training paradigm of Section 2.2. At inference, its parameters are updated online by a gradient step on a meta-learned self-supervised loss ℓ_{self} . We specify ℓ_{self} as a reconstruction objective defined through two learnable linear projections P_K and P_V , both meta-learned during training so that the resulting update improves value estimation. Concretely, with $P_K \in \mathbb{R}^{d' \times 2d}$ producing a perturbed input view and $P_V \in \mathbb{R}^{d' \times 2d}$ producing the reconstruction target, the self-supervised loss is

$$\ell_{\text{self}}(z_t; \theta_{t-1}, P_K, P_V) = \|f_{\text{adapt}}(P_K z_t; \theta_{t-1}) - P_V z_t\|^2. \quad (1)$$

where θ_{t-1} denotes the parameters of f_{adapt} prior to the update at timestep t . The parameters are then advanced by a single gradient descent step,

$$\theta_t = \theta_{t-1} - \eta \nabla_{\theta} \ell_{\text{self}}(z_t; \theta_{t-1}), \quad (2)$$

with learning rate η . Since each update is initialized from θ_{t-1} , the sequence of updates causes f_{adapt} to retain information from past observations in its parameters, thereby encoding temporal context. Section 4.6.2 provides empirical evidence that these incremental per-timestep parameter updates yield stronger value estimation than a single update aggregated at the trajectory level.

3.1.3 ZERO-SHOT VALUE FUNCTION ESTIMATOR

After the test-time update, a meta-learned linear projection $P_Q \in \mathbb{R}^{d' \times 2d}$ maps the input z_t into the adaptation space $\mathbb{R}^{d'}$. The projected representation is passed through the adaptation module f_{adapt} with the updated parameters θ_t , and then through a regression head h that emits the goal-conditioned value estimate:

$$V(z_t; g) = h(f_{\text{adapt}}(P_Q z_t; \theta_t)) \quad (3)$$

The regression head h is a two-layer multilayer perceptron (MLP), consistent with value-function architectures used in deep reinforcement learning (Espeholt et al., 2018). It is trained against the normalized progress labels y_t from expert demonstrations defined in Section 2.1, under a supervised prediction loss ℓ_{pred} equal to the mean squared error between the predictions $V(z_t; g)$ and the targets y_t . At inference, h is held frozen while only f_{adapt} is updated online via the self-supervised objective; this decoupling is what permits zero-shot generalization without task-specific demonstrations.

3.2 TRAINING PROCEDURE

We optimize the value function estimator by gradient-based meta-learning, shaping the self-supervised loss ℓ_{self} so that the test-time update it induces reduces the supervised prediction loss ℓ_{pred} (Sun

et al., 2024). Optimization is carried out over diverse sub-trajectories chosen by dissimilarity-based sampling, which suppresses shortcut learning and steers the estimator toward semantic and temporal cues.

3.2.1 META-LEARNING

During training, the adaptation module f_{adapt} is updated online at each timestep by the self-supervised loss ℓ_{self} , exactly as it will be at inference. Following the gradient-based meta-learning paradigm (Finn et al., 2017), we backpropagate through these inner adaptation updates to differentiate the supervised prediction loss ℓ_{pred} with respect to the meta-parameters: the initialization θ_0 of f_{adapt} , the linear projections P_K, P_V, P_Q , and the regression head h . The total training objective is the sum of the supervised prediction loss ℓ_{pred} and the self-supervised loss ℓ_{self} scaled by a coefficient λ . The optimized initialization θ_0 is retained as the starting point for adaptation at inference. This bilevel scheme trains the estimator to adapt its internal representations at test time in a direction that improves value estimation.

3.2.2 DISSIMILARITY-BASED SAMPLING

Consecutive frames in expert visual trajectories are typically highly redundant, a property noted in prior work on video action recognition (Wang et al., 2018). This redundancy predisposes the value function estimator to exploit shortcut cues (Geirhos et al., 2020; Lee et al., 2021b), overfitting to visual patterns that recur disproportionately in late task stages. To counter this bias, we propose a dissimilarity-based sampling strategy that assembles each mini-batch from the most visually diverse sub-trajectories of a given trajectory. The strategy raises intra-batch variance and functions as an importance-sampling scheme that up-weights semantically diverse segments.

Formally, from a sequence of multimodal representations $\{z_1, \dots, z_T\}$ we extract fixed-length sub-trajectories with a sliding window of size w_{tr} and stride s , producing a candidate set $\mathcal{W}$. We then seek the size- k subset that maximizes total pairwise dissimilarity:

$$\mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{\{w^i, w^j\} \in \binom{\mathcal{U}}{2}} \|w^i - w^j\|_2^2, \quad (4)$$

where each w^i denotes a sub-trajectory of length w_{tr} . Exact optimization of Eq. 4 is combinatorially intractable, as it enumerates all $\binom{|\mathcal{W}|}{k}$ subsets. We therefore approximate it with a scoring-based heuristic that assigns each window $w \in \mathcal{W}$ a score equal to its summed dissimilarity to all other windows:

$$s(w) = \sum_{v \in \mathcal{W}} \|w - v\|_2^2, \quad \mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{w \in \mathcal{U}} s(w). \quad (5)$$

Selecting the k windows with the largest scores s thus yields a heuristically diverse subset in polynomial time. As quantified by the complexity analysis in Appendix G, this selection contributes negligible overhead relative to model training. The ablation in Section 4.6.1 empirically contrasts dissimilarity-based sampling with full-trajectory sampling, showing that the former improves discrimination of expert from non-expert visual trajectories.

4 EXPERIMENTS

We assess VITA against three core capabilities required of a goal-conditioned value function: (i) generalization of value estimation on real-world robotic manipulation under distribution shifts in task, environment, and embodiment; (ii) discrimination between expert and non-expert visual trajectories on real-world robotic manipulation tasks; (iii) zero-shot dense reward shaping for offline RL on Meta-World MT10, a simulated multi-task robot-learning benchmark.

4.1 TRAINING SETUP

We train VITA on expert visual trajectories paired with natural-language task descriptions drawn from BridgeData V2 (Walke et al., 2023), a dataset spanning a broad range of manipulation tasks, environments, and robot embodiments. Specifically, the training set is a curated subset of 2,986 expert

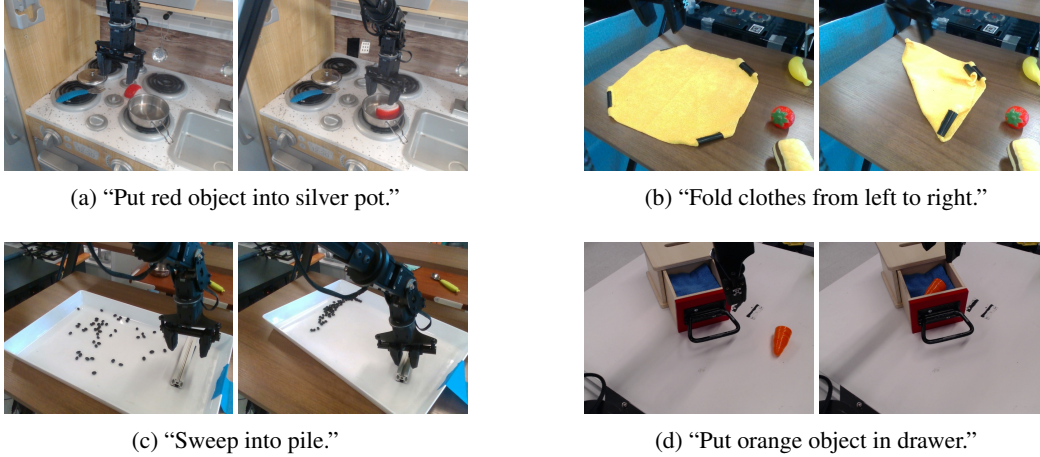


Figure 3: Representative visual trajectories with their task descriptions, grouped by distribution-shift regime. (a) In-distribution. (b, c) Environment shift. (d) Joint embodiment-and-environment shift.

demonstrations restricted to pick-and-place manipulation; it deliberately excludes folding, sweeping, and stacking tasks. All demonstrations are recorded with a single embodiment (WidowX 250) across four configurations of the ToyKitchen environment. A disjoint set of 287 expert demonstrations is held out as the in-distribution test set, denoted `tk_pnp`; Appendix A illustrates training examples. The frozen backbone for encoding video frames and task descriptions is OpenCLIP ViT-B/32. The training objective is the weighted sum of ℓ_{pred} and ℓ_{self} , with self-supervised weight $\lambda_{\text{self}} = 0.5$. Dissimilarity-based sampling is configured with window size $w_{\text{tr}} = 8$, number of sub-trajectories $k = 8$, and stride $s = 1$. At test time, adaptation uses a single gradient step ($t_{\text{ep}} = 1$) with λ_{self} and learning rate $\eta = 0.1$. Full details and hyperparameter sweeps are given in Appendix D.

4.2 BASELINES

We benchmark VITA against value functions derived from both contrastive and autoregressive VLMs. VLM-CL produces zero-shot value estimates from the cosine similarity between the CLIP embedding of each frame and that of the task description (Baumli et al., 2023). VLM-RM regularizes the CLIP embedding by projecting it onto the direction from a generic reference prompt to the task prompt (Rocamonde et al., 2024). CLIP-FT fits a supervised regression head on frozen CLIP multimodal representations, whereas CLIP-GRU fits a Gated Recurrent Unit (GRU) (Cho et al., 2014) on the same representations. GVL (Ma et al., 2024) is the state-of-the-art zero-shot value function built on autoregressive VLMs; we evaluate two of its regimes. GVL-0S is the zero-shot regime, in which the VLM is prompted with only the test trajectory and task description, while GVL-1S is the one-shot in-context regime, in which a single shuffled training-distribution trajectory and its progress labels are supplied as an exemplar. Consistent with GVL, we use Gemini 1.5 Pro (`gemini-1.5 Pro-latest`) (Gemini Team et al., 2024) with its proposed prompt template. We additionally evaluated Qwen-VL 2.5 (Yang et al., 2024), which did not overcome the monotonic temporal bias, and GPT-4o (Hurst et al., 2024), which declined to emit scalar value estimates. Further implementation details appear in Appendix D.

4.3 EVALUATING GENERALIZATION UNDER DISTRIBUTION SHIFTS

We quantify how well progress estimators generalize across unseen environments, tasks, and robot embodiments, using subsets of BridgeData V2 curated to isolate variation along each axis. Environment shifts alter scene layout or background: `lm_pnp` is a pick-and-place task performed in front of a laundry machine, `td_fold`, `ft_fold`, and `rd_fold` are cloth-folding tasks on distinct surfaces, and `ms_sweep` is a long-horizon sweeping task confined to a tray. Embodiment shifts are induced with the DeepThought robot, which differs from the training embodiment (WidowX 250) in both morphology and camera viewpoint. Within the embodiment-shift group, `dt_tk_pnp` (pick-and-place) and `dt_tk_stack` (stacking) preserve the in-distribution environment while changing

Table 1: Value Order Correlation (VOC) for value function estimation under distribution shift, reported per evaluation dataset. Regimes: **ID** = In-Distribution, **ES** = Environment Shift, **EM** = Embodiment Shift, **ES & EM** = joint environment-and-embodiment shift. Best result per row in **bold**, second-best underlined.

Shift	Dataset	VLM-CL	VLM-RM	CLIP-FT	GVL-0S	GVL-1S	CLIP-GRU	VITA
ID	tk_pnp	0.038	0.029	0.251	0.269	0.252	<u>0.773</u>	0.782
ES	lm_pnp	0.017	0.033	0.149	0.305	0.272	<u>0.676</u>	0.725
	td_fold	0.031	0.072	0.152	0.326	0.318	<u>0.674</u>	0.709
	ft_fold	0.108	0.099	0.162	0.331	0.387	0.693	<u>0.658</u>
	rd_fold	0.095	0.055	0.126	0.372	0.406	0.726	<u>0.606</u>
	ms_sweep	-0.129	-0.226	0.148	0.158	0.150	<u>0.434</u>	0.490
EM	dt_tk_pnp	0.042	-0.041	0.149	0.258	0.211	0.856	<u>0.820</u>
	dt_tk_stack	0.035	0.046	0.099	0.254	0.277	<u>0.667</u>	0.708
ES & EM	dt_ft_stack	0.026	0.028	0.049	0.212	0.249	<u>0.674</u>	0.698
	dt_rd_pnp	0.023	0.041	0.211	0.329	0.316	0.747	<u>0.695</u>

only the embodiment, whereas `dt_ft_stack` (stacking) and `dt_rd_pnp` (drawer pick-and-place) impose both embodiment and environment shifts. Each dataset comprises 200 expert trajectories with task descriptions, except `ms_sweep`, which contains 100 owing to data availability. Appendix B fully specifies the evaluation datasets. Performance is measured by the *Value Order Correlation* (VOC) metric (Ma et al., 2024), which quantifies the agreement between predicted progress and the chronological frame ordering of a trajectory. Formally, letting $p_1, \dots, p_T$ denote the predicted progress values for a T -frame trajectory and $r_t = t$ the temporal index, VOC is the Spearman rank correlation ρ_s between the predicted values and the temporal indices.

Table 1 shows that VITA and CLIP-GRU both dominate GVL-0S and GVL-1S across the board, which quantifies the limitation of pre-trained autoregressive VLMs and confirms that preserving temporal order is decisive for progress estimation. Among the two temporally aware methods, VITA attains the higher VOC on a majority of the evaluation datasets (6 of 10). Both further exceed the CLIP-based baselines that omit temporal modeling (CLIP-FT, VLM-CL, VLM-RM), isolating the contribution of encoding temporal history. CLIP-FT matches GVL in-distribution but does not transfer to out-of-distribution settings, and VLM-CL and VLM-RM perform poorly throughout, consistent with their lack of temporal modeling. For GVL-0S versus GVL-1S, the in-context exemplar helps on some datasets, is neutral on others, and is occasionally detrimental. Both GVL variants perform well on folding tasks (`td_fold`, `ft_fold`, `rd_fold`) yet degrade substantially on stacking (`dt_tk_stack`, `dt_ft_stack`) and pick-and-place (`dt_tk_pnp`, `lm_pnp`), indicating that the underlying autoregressive VLM is biased toward folding-like manipulations. VITA, by contrast, maintains consistent VOC across all task types and shift regimes, evidencing stronger generalization than in-context learning.

On the long-horizon task `ms_sweep`, every method except CLIP-FT degrades, yet VITA still attains the highest VOC and, in particular, surpasses CLIP-GRU—evidence that for long-horizon manipulation, encoding temporal history in adapter parameters via test-time adaptation is more effective than encoding it in recurrent hidden states. VITA also generalizes strongly under embodiment shift. On `dt_tk_pnp`, which changes only the embodiment while retaining the training environment and tasks, both VITA and CLIP-GRU exceed their in-distribution VOC, indicating that learned temporal context transfers across embodiments. GVL-1S, conversely, performs poorly on both `dt_tk_pnp` and `dt_tk_stack`, indicating that one-shot in-context learning does not generalize under embodiment shift in our evaluation setup.

4.4 DIFFERENTIATING EXPERT FROM NON-EXPERT TRAJECTORIES

Beyond generalization, we probe the robustness of VITA through its capacity to separate expert from suboptimal visual trajectories by assigning the latter lower progress scores. We contrast model predictions on expert versus scripted (non-expert) trajectories drawn from the same in-distribution setting of BridgeData V2. The scripted trajectories are produced in the ToyKitchen environment by a heavily

randomized controller (Walke et al., 2023), holding the embodiment and environment configuration fixed to those of the expert demonstrations. They comprise pick-and-place tasks over object categories that may overlap with the training set: general objects (`sc_pnp_obj`), rigid items (`sc_pnp_robj`), soft toys (`sc_pnp_stoy`), utensils (`sc_pnp_uten`), and a cloth (`sc_pnp_cloth`). Although these trajectories match the training distribution in task, embodiment, and environment, their behavior departs from the smooth, monotonic progress characteristic of expert demonstrations. Appendix C shows scripted examples. Because the randomized controller does not target a quantifiable level of suboptimality, we do not assume calibrated degradation; we require only that an effective estimator assign lower VOC to suboptimal than to expert trajectories evaluated in the same setting. We therefore measure discriminative success by *BinVOC*, the binary indicator $\mathbf{1}[\text{VOC}_{\text{exp}} > \text{VOC}_{\text{subopt}}]$, equal to 1 exactly when the expert trajectory attains higher VOC than the suboptimal one and 0 otherwise.

Discriminative performance is reported as *BINVOC* averaged over all scripted datasets, with expert performance (VOC_{exp}) computed on the in-distribution test set `tk_pnp`, whose pick-and-place tasks parallel those in the scripted evaluation. VITA, GVL-0S, and GVL-1S achieve perfect discrimination, assigning higher VOC to the expert trajectory on every task. VITA exceeds CLIP-GRU, indicating that storing temporal memory in recurrent hidden states is more susceptible to overfitting temporal shortcuts than accumulating an implicit memory through sequential test-time updates. The CLIP-based baselines (VLM-CL, VLM-RM) underperform, consistent with their absence of temporal modeling, while CLIP-FT is more reliable but fails on one task, indicating sensitivity to shifts in the object distribution.

4.5 ZERO-SHOT REWARD SHAPING FOR OFFLINE RL

We evaluate the value function estimator on the Meta-World MT10 benchmark (Yu et al., 2020), comprising ten distinct robotic manipulation tasks (e.g., pick-and-place, door opening, pushing), in order to assess VITA as a zero-shot reward-shaping signal for offline RL. VITA is trained on real-world robotic data and applied zero-shot in this simulated domain. For each MT10 task we roll out 20 expert visual demonstrations from Meta-World’s expert policies and use the goal-conditioned value estimator to assign a dense reward to every visual observation. Policies are then trained by Implicit Q-Learning (IQL) (Kostrikov et al., 2021) in the offline regime. We repeat demonstration generation and policy training over 10 random seeds; each resulting multi-task policy is evaluated on 20 rollouts per task, and we record the mean episodic return per (task, seed) pair. Following the protocol of Agarwal et al. (2021), we report the interquartile mean (IQM) over all pairs with 95% stratified bootstrap confidence intervals. Training details and the full evaluation protocol are given in Appendix E.

Table 2 reports MT10 performance for all baselines under this protocol. VITA attains the highest IQM return, 0.815, exceeding every CLIP-based baseline. A direct comparison with GVL is infeasible at this scale: GVL relies on Gemini-1.5, a proprietary autoregressive VLM whose per-query inference cost is prohibitive in the RL loop, and the open-source alternatives we tried did not yield reliable progress estimates in preliminary tests. As a reference point, Meta-World’s own fuzzy-logic dense reward (META-WL) attains 0.779. VITA exceeds this strong reference, indicating that a value estimator trained on real-world data can transfer effectively to reward shaping in simulation.

4.6 ABLATION STUDIES

4.6.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

We isolate the contribution of dissimilarity-based sampling by varying only the training-time sub-trajectory sampling strategy and measuring the effect on VITA’s expert-versus-non-expert discrimination. Full-trajectory sampling (Sun et al., 2024) evaluates the adaptation loss over all overlapping sub-trajectories at stride $s = 1$; in our setup this induces overfitting to global temporal shortcuts. Uniform random sampling of sub-trajectories increases coverage but does not control for semantic diversity. Dissimilarity-based sampling instead selects the sub-trajectories that maximize pairwise dissimilarity, explicitly promoting semantic diversity, and it outperforms both full-trajectory and random sampling on expert–non-expert discrimination. Appendix F.1 reports the full comparison.

Method	BinVOC
VLM-CL	0.40
VLM-RM	0.00
CLIP-FT	0.80
GVL-0S	1.00
GVL-1S	1.00
CLIP-GRU	0.80
VITA	1.00

(a) Expert vs. Non-Expert (BinVOC).

Method	IQM	95% CI
VLM-CL	0.760	[0.722, 0.791]
VLM-RM	0.746	[0.718, 0.771]
CLIP-FT	0.785	[0.759, 0.809]
CLIP-GRU	0.777	[0.734, 0.814]
META-WL	0.779	[0.750, 0.804]
VITA	0.815	[0.785, 0.838]

(b) Offline RL on MT10 (IQM).

Table 2: (a) Discrimination of expert from scripted robot demonstrations, measured by BinVOC averaged over the 5 in-distribution scripted datasets. (b) Offline RL performance on Meta-World MT10, measured by interquartile mean (IQM) of return with 95% stratified bootstrap confidence intervals.

4.6.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

We analyze how the choice of test-time adaptation strategy for incorporating temporal context affects value function estimation. VITA performs per-timestep adaptation without reset (implicit memory), advancing the parameters incrementally from θ_{t-1} ; this encodes trajectory history directly in the parameters, capturing both temporal order and accumulated context. We compare VITA against three ablated variants: (i) *trajectory-level adaptation (TTT-TR)*, a single gradient update computed over the entire trajectory—a batched update that averages the loss across all frames and thereby discards temporal order; (ii) *memoryless adaptation (TTT-RS)*, which resets the module to θ_0 at each step and adapts to the current observation alone, carrying no temporal information forward; (iii) *explicit memory (TTT-EX)*, which likewise resets the module at each step but adapts over a local window of recent frames matching the meta-training window size ($w_{tr} = 8$). Across all datasets, VITA consistently outperforms the trajectory-level, memoryless, and explicit-memory variants, establishing that incremental accumulation of temporal context is the effective mechanism for value function estimation. The full analysis is given in Appendix F.2.

5 RELATED WORK

5.1 VLMS AS VALUE FUNCTION ESTIMATORS

Contrastive VLMS have been deployed as zero-shot reward models (Baumli et al., 2023; Rocamonde et al., 2024) and goal-conditioned value functions (Ma et al., 2023b; 2024), scoring progress by frame-level similarity but omitting temporal modeling. A separate strand fine-tunes CLIP on domain-specific video corpora (Fan et al., 2022; Jiang et al., 2024) to emit dense rewards, at the cost of supervision that curtails zero-shot applicability. RoboCLIP (Sontakke et al., 2023) supports one-shot reward specification without fine-tuning, but produces only sparse rewards and still requires a demonstration. Large-scale multimodal pre-training methods (Alakuijala et al., 2024; Li et al., 2024; Ma et al., 2023b) learn language-conditioned value functions that rely on data scale for generalization, and ReWiND (Zhang et al., 2025b) trains a cross-modal sequential transformer on augmented large-scale data with video rewinding, augmented by a small set of in-domain demonstrations. VITA departs from all of these: it adapts a frozen contrastive vision–language encoder at inference, capturing temporal and semantic context through test-time adaptation, and therefore requires neither large-scale multimodal pre-training nor additional domain-specific demonstrations to generalize. Because the mechanism is agnostic to the encoder, VITA can be layered on any pre-trained vision–language representation that encodes a universal value function, augmenting its generalization and temporal reasoning through meta-learned test-time adaptation. On the autoregressive side, VLMS (Alayrac et al., 2022) exploit in-context learning over trajectories and have been applied to success detection (Du et al., 2023) and goal-conditioned value functions (Ma et al., 2024). GVL (Ma et al., 2024) counters the monotonic bias of ordered pre-training data by shuffling frames at inference, but this discards the temporal order that temporal reasoning—and the disambiguation of visually similar yet temporally distinct states—depends on. VITA instead preserves chronological order and recovers temporal context through test-time adaptation, avoiding this shortcut.

5.2 TEST-TIME ADAPTATION

Test-time adaptation methods adapt VLMs parameter-efficiently, without full fine-tuning. Prompt-based methods optimize a small set of parameters at test time from the task context, though the resulting mechanism is largely opaque (Zhou et al., 2022; Khattak et al., 2023; Ma et al., 2023a). Lim et al. (2025) argue that CLIP representations are amenable to test-time adaptation, on the grounds that CLIP encodes human-interpretable concepts surfaced through mechanistic interpretability (Bricken et al., 2023). CLIP similarity has additionally been used as a reward whose parameters are optimized at test time by reinforcement learning (Zhao et al., 2024). Test-time training (Sun et al., 2020) constitutes a further approach, updating an adaptation module at each timestep via a self-supervised loss. In contrast to meta-reinforcement learning (Finn et al., 2017; Bauer et al., 2023), which requires task-specific adaptation episodes at training time, test-time training performs online adaptation directly at inference without task labels.

6 DISCUSSION, LIMITATIONS, AND FUTURE WORK

6.1 DISCUSSION

Our results establish that test-time adaptation lets value functions for robotic manipulation generalize across distribution shifts in task, environment, and embodiment. VITA additionally discriminates expert from non-expert visual trajectories and supplies a zero-shot reward-shaping signal, which enables downstream use in RL and imitation learning. Because it updates sequentially along a trajectory, VITA encodes history in its parameters and thereby captures temporal and semantic context more effectively than the CLIP-based baselines and than in-context learning with autoregressive VLMs. The margin of VITA over CLIP-GRU on both expert–non-expert discrimination and offline RL indicates that, for zero-shot value estimation, an implicit memory formed by sequential test-time updates generalizes better than temporal history held explicitly in recurrent hidden states. Pre-trained autoregressive VLMs such as GVL, though broadly capable, are not optimized for progress estimation, a task that demands temporal reasoning over visual trajectories. Finally, the inference-time cost of per-timestep adaptation is negligible in our setup, since only a lightweight module is updated, so real-time applicability is preserved.

6.2 LIMITATIONS AND FUTURE WORK

In our experiments test-time adaptation improves the generalization of zero-shot progress estimation to unseen tasks and environments, yet it remains challenged by settings with high execution variability or long durations. Although the adaptation overhead is negligible given the lightweight module, updating the value function estimator at every timestep may be unsafe under deployment, which constrains applicability in real-time scenarios. Future work will investigate alternative test-time adaptation schemes for vision–language models, particularly for training agents inside world models; applying VITA to real-time closed-loop control and to more complex RL environments is a further direction. Since the mechanism is agnostic to the underlying vision–language encoder, VITA could augment the generalization and temporal reasoning of any domain-specific pre-trained multimodal representation through meta-learned test-time adaptation. Finally, while we provide empirical evidence that dissimilarity-based sampling mitigates shortcut learning, a formal analysis of how diversity-based sampling shapes shortcut learning remains open; it lies beyond our core contribution, which is a value function estimator that meta-learns a test-time adaptation mechanism to improve zero-shot performance and temporal reasoning.

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A TRAINING DATASET

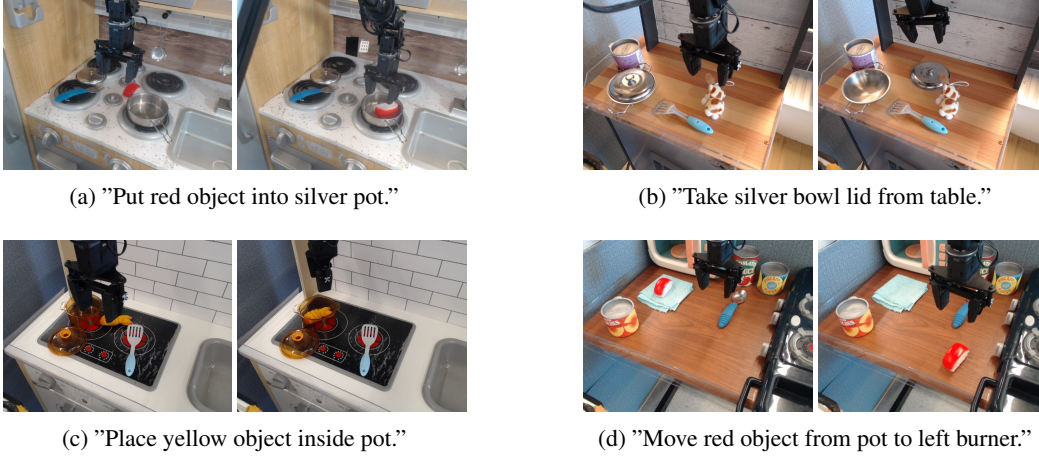


Figure 4: Each subfigure pairs the start and end frames of an expert training demonstration with its natural-language task description. Demonstrations span four distinct ToyKitchen environment configurations.

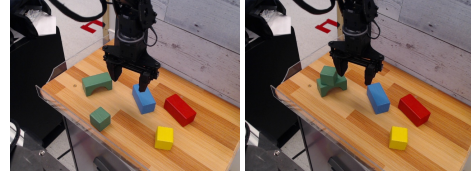
B EVALUATION DATASETS

Table 3: Specification of each evaluation dataset by task type, environment, and robot embodiment.

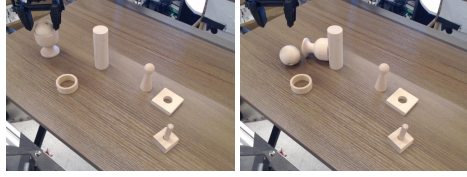
Dataset	Task Type	Environment	Embodiment
tk_pnp	pick-and-place	toy kitchen	WidowX 250
lm_pnp	pick-and-place	laundry machine	WidowX 250
td_fold	fold cloth	tabletop (dark wood)	WidowX 250
ft_fold	fold cloth	folding table	WidowX 250
rd_fold	fold cloth	robot desk	WidowX 250
ms_sweep	sweep	folding table (tray)	WidowX 250
dt_tk_pnp	pick-and-place	toy kitchen	DeepThought
dt_tk_stack	stack blocks	toy kitchen	DeepThought
dt_ft_stack	stack blocks	folding table	DeepThought
dt_rd_pnp	pick-and-place	robot desk (drawer)	DeepThought



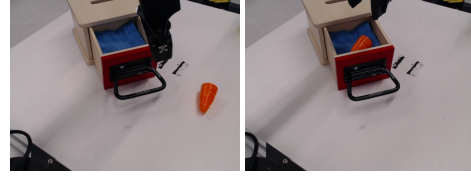
(a) dt_tk_ms: "Move yellow object from burner to center."



(b) dt_tk_stack: "Place green cube on top of arch."

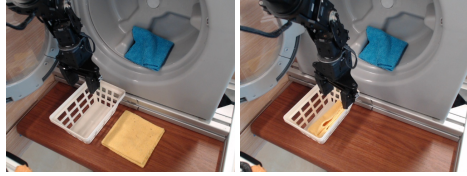


(c) dt_ft_stack: "Move egg to table."



(d) dt_rd_pnp: "Put orange object in drawer."

Figure 5: Each subfigure pairs the start and end frames of an evaluation trajectory under embodiment shift with its natural-language task description. The top row holds the environment fixed (ToyKitchen) while changing the robot to DeepThought; the bottom row additionally changes the environment.



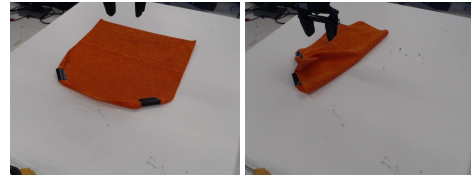
(a) lm_pnp: "Place blue cloth inside washer."



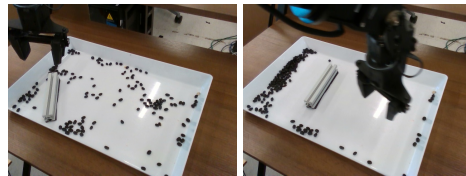
(b) td_fold: "Fold cloth bottom right to top left."



(c) ft_fcnp: "Fold clothes from left to right."



(d) rd_fold: "Fold cloth bottom right to top left."



(e) ms_ft_sweep: "Sweep into pile."

Figure 6: Each subfigure pairs the start and end frames of an evaluation trajectory under environment shift with its natural-language task description. The embodiment is held fixed while the environment varies visually and structurally.

C SCRIPTED DATASET



(a) `sc_pnp_obj`: "Pick up an object and place it."



(b) `sc_pnp_robj`: "Pick up a rigid object and place it."



(c) `sc_pnp_stoy`: "Pick up a soft object and place it."



(d) `sc_pnp_uten`: "Pick up a kitchen utensil and place it."



(e) `sc_pnp_cloth`: "Move the cloth."

Figure 7: Each subfigure pairs the start and end frames of a scripted evaluation trajectory, collected in the ToyKitchen environment with the WidowX 250 robot. All tasks are pick-and-place over varying object categories; we assign generic task descriptions in the absence of publicly available annotations.

D MODEL ARCHITECTURE

D.1 MODEL AND TRAINING DETAILS

The frozen backbone is the OpenCLIP ViT-B/32 encoder pre-trained on the OpenAI dataset. Each visual observation and each task description is embedded by the CLIP vision and text encoders, respectively. We adopt a joint (concatenation-based) representation rather than an element-wise product, a choice determined by superior validation performance. The adaptation module f_{adapt} is a two-layer residual MLP with GELU activations and projection dimension $d' = 64$. Optimization uses AdamW with learning rate 1×10^{-4} , weight decay 1×10^{-4} , and a cosine schedule with 10% warmup, at batch size 32; all trajectories are padded to the maximum training-set length of 120 frames. The self-supervised weight in the total objective is fixed at $\lambda_{\text{self}} = 0.5$, selected on validation performance. We train for 5 epochs, since validation VOC plateaus early and longer training yields no further gain. Unless stated otherwise, dissimilarity-based sampling uses stride $s = 1$, sub-trajectory length $w_{\text{tr}} = 8$, and $k = 8$ selected windows. All experiments were run on NVIDIA RTX 6000 Ada Generation GPUs.

D.2 TEST-TIME TRAINING HYPERPARAMETERS

At inference, only the adaptation module f_{adapt} is updated, retaining the training projection dimension $d' = 64$, via a single gradient step ($t_{\text{ep}} = 1$) at learning rate $\eta = 0.1$. This configuration was chosen by a grid sweep over learning rates $\{5.0, 1.0, 0.1, 0.05, 0.01, 0.001\}$ and adaptation-step counts $\{1, 5, 10\}$, scored on validation performance in Table 4. Two configurations attain comparable validation VOC: (i) one step at the larger rate ($\eta = 0.1$, $t_{\text{ep}} = 1$) and (ii) ten steps at the smaller

Table 4: Validation VOC of VITA as a function of the test-time learning rate η and the number of adaptation steps t_{ep} .

η	$t_{\text{ep}} = 1$	$t_{\text{ep}} = 5$	$t_{\text{ep}} = 10$
0.001	0.225	0.345	0.453
0.01	0.454	0.744	0.781
0.05	0.746	0.670	0.515
0.1	0.783	0.512	0.369
1.0	0.299	0.183	0.181
5.0	0.169	0.182	0.157

rate ($\eta = 0.01$, $t_{\text{ep}} = 10$). We adopt the former, as it requires fewer adaptation steps while scoring marginally higher.

D.3 BASELINES

VITA uses $t_{\text{ep}} = 1$, projection dimension $d' = 64$, and learning rate $\eta = 0.1$, fixed by the hyperparameter selection described above. CLIP-FT shares the VITA architecture but omits the adaptation module and the self-supervised loss: it applies a frozen CLIP encoder followed by a linear projection and a two-layer MLP to regress task progress, trained by standard supervised regression without meta-learning or test-time adaptation. To match expressivity, it is given an $8\times$ larger projection matrix and $10\times$ more training steps than our method (Lin et al., 2022). For VLM-RM, the reference prompt is a baseline description of the environment. CLIP-GRU likewise inherits the VITA architecture (frozen CLIP encoder and regression head h) but substitutes a Gated Recurrent Unit (GRU) for the test-time adaptation module f_{adapt} , encoding temporal history explicitly in recurrent hidden states; the GRU is single-layer with hidden size 64, and its final hidden state feeds h . For both GVL-0S and GVL-1S we use the latest Gemini 1.5 Pro release, `gemini-1.5-pro-latest` (Gemini Team et al., 2024), whereas the original GVL implementation used the earlier `gemini-1.5-pro`. We additionally evaluated GVL with the open-source autoregressive VLM `Qwen-VL 2.5` (Yang et al., 2024), which did not overcome the temporal bias despite frame shuffling, and with GPT-4o (Hurst et al., 2024), which consistently declined to emit scalar progress estimates in our setup. All GVL evaluations followed the prompt template of the original work (Ma et al., 2024).

E OFFLINE RL SETUP

E.1 TRAINING CONFIGURATION

We instantiate Implicit Q-Learning (IQL) with the following hyperparameters, held fixed across all tasks: expectile $\tau = 0.7$, advantage-weighting temperature 3.0, and weight-clipping threshold 100.0. The actor and critic use learning rates 1×10^{-4} and 3×10^{-4} , respectively, at batch size 256. Each policy is trained for 100,000 gradient steps and evaluated every 10,000 steps on 20 rollouts per task at horizon 150. The implementation follows `d3rlpy` with configuration `expectile=0.7`, `weight.temp=3.0`, `max.weight=100.0`, `actor.lr=1e-4`, `critic.lr=3e-4`, `batch.size=256`.

E.2 EVALUATION PROTOCOL

Following Agarwal et al. (Agarwal et al., 2021), we report the interquartile mean (IQM) of returns over all task–seed pairs. The IQM averages the middle 50% of scores and is thus less sensitive to outliers than either the mean or the median. We estimate 95% confidence intervals by stratified bootstrap with 10,000 resamples, stratified by task. This procedure is applied identically to all methods in the Meta-World MT10 experiments.

F ABLATION STUDIES

F.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

Table 5: Disc@VOC across scripted datasets under different training-time sampling strategies for test-time adaptation. A checkmark denotes successful discrimination of expert from suboptimal trajectories. The sampling strategies **FT** (full-trajectory), **Rand** (random), and **Diss** (dissimilarity-based) are parameterized by (w_{tr}, k) .

Dataset	FT (8,-)	Rand (8,8)	Diss (8,4)	Diss (4,8)	Diss (8,8)
sc_pnp_obj		✓	✓		✓
sc_pnp_robj	✓	✓	✓	✓	✓
sc_pnp_stoy			✓		✓
sc_pnp_uten	✓	✓	✓	✓	✓
sc_pnp_cloth		✓	✓	✓	✓
Avg. Disc@VOC	0.20	0.8	1.0	0.60	1.0

We validate dissimilarity-based sampling by quantifying how the training-time sub-trajectory sampling strategy affects the model’s ability to discriminate expert from suboptimal trajectories. In Table 5 we compute Disc@VOC over 5 scripted datasets, where Disc@VOC records whether the model assigns a higher average VOC to the expert set (`tk_pnp`) than to a given scripted dataset. We contrast three strategies: (1) full-trajectory sampling, which adapts over all overlapping length- w_{tr} sub-trajectories at stride s , following prior work (Sun et al., 2024); (2) random sampling, which draws k length- w_{tr} sub-trajectories uniformly per video; and (3) our dissimilarity-based sampling, which forms a candidate set of length- w_{tr} sub-trajectories at stride s and selects the k that maximize pairwise dissimilarity in the representation space.

Dissimilarity-based sampling with $w_{tr} = 8$ and $k = 8$ achieves perfect discrimination. Reducing the number of windows to $k = 4$ at $w_{tr} = 8$ preserves this performance, whereas shortening the sub-trajectory length to $w_{tr} = 4$ degrades it. The full-trajectory baseline is worst, confirming that adapting over all consecutive windows overfits to global temporal cues; random sampling improves on it but remains less reliable than dissimilarity-based selection. Taken together, these results indicate that both sub-trajectory diversity and semantic locality are necessary for discriminative test-time adaptation.

F.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

Table 6: Validation VOC for progress estimation under distribution shift, per dataset and adaptation variant. ID = In-Distribution, ES = Environment Shift, EM = Embodiment Shift, ES+EM = joint environment-and-embodiment shift.

Shift	Dataset	TTT-TR	TTT-RS	TTT-EX	VITA
ID	tk_pnp	0.1917	0.1918	0.1957	0.7822
ES	lm_pnp	0.1843	0.1854	0.1826	0.7246
	td_fold	0.1402	0.1393	0.1322	0.7085
	ft_fold	0.1302	0.1311	0.1683	0.6583
	rd_fold	0.1161	0.1172	0.1154	0.6056
	ms_sweep	-0.0225	-0.0191	0.0102	0.4898
EM	dt_tk_pnp	0.2123	0.2118	0.2069	0.8203
	dt_tk_stack	0.0833	0.0830	0.0747	0.7081
ES+EM	dt_ft_stack	0.0558	0.0537	0.0439	0.6979
	dt_rd_pnp	0.1665	0.1660	0.1559	0.6951

We evaluate four adaptation strategies that differ only in how temporal context is incorporated: (1) TTT-TR, a single offline gradient update over the full trajectory; this batched update averages the loss across all frames and hence discards temporal order; (2) TTT-RS, which resets the module at every step and updates from the current observation alone (memoryless), carrying no temporal information forward; (3) TTT-EX, which resets the module at every step but updates over a local window of recent frames matching the meta-training window size ($w_{\text{tr}} = 8$), averaging gradients within the window and thereby discarding intra-window order; (4) VITA (OURS), which updates the module incrementally at every timestep (implicit memory), sequentially accumulating temporal history in the adapter parameters. We ran the identical test-time hyperparameter sweep (Appendix D) for all variants. VITA, TTT-RS, and TTT-TR are optimal at learning rate 0.1 with one adaptation step, whereas TTT-EX is optimal at learning rate 1.0 with one adaptation step.

Table 6 shows that TTT-TR, TTT-RS, and TTT-EX perform comparably: batched updates, whether over a short explicit window (TTT-EX) or the full trajectory (TTT-TR), are on par with the memoryless baseline (TTT-RS). VITA consistently exceeds all three, which establishes that sequential, per-timestep updates are the operative mechanism for value estimation on real-world robotic manipulation. Two properties explain why the implicit-memory paradigm demands exactly this update schedule. First, under implicit memory, updating over a local window would double-count the most recent frames—once explicitly through the window and once implicitly through the accumulated parameters. Second, updating less frequently (non-sequentially) would omit intermediate frames and lose their information. Consequently, the implicit-memory test-time training paradigm requires sequential, per-timestep updates without reset.

G COMPLEXITY ANALYSIS OF DISSIMILARITY-BASED SAMPLING

Let $\mathcal{W}$ be the candidate set with $N_c = |\mathcal{W}|$. To approximate diverse-subset selection without the exponential cost of enumerating all $\binom{N_c}{b}$ subsets in Eq. 4, our scoring-based heuristic assigns each candidate window a diversity score equal to the sum of its pairwise distances to all others—equivalently, the corresponding row sum of the distance matrix. The number of candidate windows is $N_c = T - w_{\text{tr}} + 1$, each represented by a flattened feature vector of dimension $m = w_{\text{tr}} \cdot d$, where d is the dimension of a multimodal representation z_t . The full pairwise distance matrix $D \in \mathbb{R}^{N_c \times N_c}$ is computed on GPU by batched `cdist` at cost $\mathcal{O}(N_c^2 \cdot w_{\text{tr}} \cdot d)$.

In our setting the training trajectories are short (mean length 69, maximum 120). With batch size 32, feature dimension $d = 1024$ (the CLIP-based multimodal representation), and window length $w_{\text{tr}} = 8$, the worst-case cost is

$$32 \cdot N_c^2 \cdot w_{\text{tr}} \cdot d = 32 \cdot 114^2 \cdot 8 \cdot 1024 \approx 340 \text{ MFLOPs.}$$

This overhead is negligible relative to a single forward–backward pass through the model, while yielding the desired selection of diverse windows.